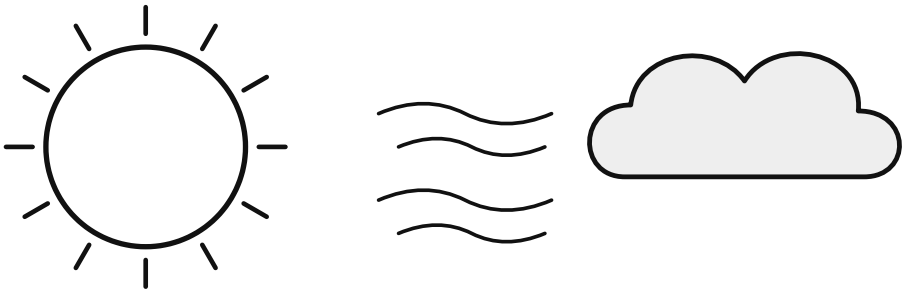
EARTH AND SPACE

# Resources and Sustainability

Renewable and non-renewable resources, regeneration time, the 5 Rs and how people affect the Earth



**In this pack**

**Learn it:** Using the Earth's resources wisely ★☆☆
**Fact check:** Renewable or not? ★☆☆
**Think it through:** Is it sustainable? ★★☆
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**Real-world case study:** Saving the hilsa ★★★
**Review and challenge:** Show what you know ★★★

**This pack belongs to:** _____ **Class:** _____

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LEARN IT · B5 · RESOURCES

# Using the Earth's resources wisely

★☆☆ Grade 5

Name: Class: Date:

**Learning goals**

- Explain what "renewable" and "non-renewable" mean.
- Compare how long different resources take to regenerate.
- Explain how human actions such as recycling, conservation and over-use affect the Earth.

Everything we use comes from the Earth: the air we breathe, the water we drink, the rice we eat, the gas that cooks it, the bricks of our homes and the phone in our pocket. These useful things are **natural resources**. Some will run out; others can last for ever if we use them with care.

## Renewable and non-renewable

A **renewable** resource is replaced by nature in a short time, or never runs out: sunlight, wind, flowing water, and living things such as trees, fish and crops, as long as we do not take too many. A **non-renewable** resource takes so long to form that, once used, it is gone for us: **coal, oil, natural gas** and metals in rocks.

## How long does it take to come back?

| Resource | Time to regenerate (approximate) |
|---|---|
| sunlight, wind | never runs out |
| hilsa fish population | 1–2 years if young fish are protected |
| bamboo | 3–5 years |
| a teak or mahogany tree | 20–50 years |
| groundwater | years to thousands of years |
| a few centimetres of soil | hundreds of years |
| coal, oil, natural gas | millions of years |

Even renewable resources can run out if we use them faster than they regenerate. Cutting a forest in one year that took 50 years to grow is not sustainable.

## Sustainability

**Sustainable** use means meeting our needs **today** without stopping people in the **future** from meeting theirs. It means using renewable resources no faster than they come back, and using non-renewable ones carefully.

**Bangladesh connection**

Bangladesh makes most of its electricity from **natural gas**, but its gas fields are slowly running out. Millions of homes in villages now use **solar panels**, and the country is building more solar parks.

## The 5 Rs

- **Refuse** what you do not need (say no to a plastic bag).
- **Reduce** how much you use.
- **Reuse** things again and again.
- **Recycle** materials into new products.
- **Rot**: compost food and plant waste.

## People can help or harm

- **Harm:** over-fishing, cutting forests, wasting water, dumping plastic, burning too much fuel.
- **Help:** recycling, tree planting, protecting young fish, national parks and saving endangered animals.

**Bangladesh first**

In **2002** Bangladesh became one of the first countries in the world to **ban thin plastic shopping bags**, after they blocked drains and made floods in Dhaka worse.

**Did you know?**

Vultures clean up dead animals and stop diseases spreading. Bangladesh lost most of its vultures because of a cattle medicine (diclofenac) that poisoned them. The medicine was banned in 2010 and "vulture safe zones" were created to help them recover.

**Key words**

| | |
|---|---|
| **natural resource** | something useful we get from nature |
| **renewable** | replaced naturally in a short time |
| **non-renewable** | will run out because it takes millions of years to form |
| **sustainable** | meeting today's needs without harming the future |
| **regenerate** | grow back or be replaced |
| **fossil fuel** | coal, oil or gas formed from ancient living things |
| **conservation** | protecting nature and resources |

FACT CHECK · B5 · RESOURCES

# Renewable or not?

★☆☆
Grade 5

Name: Class: Date:

**Learning goals**

- Classify resources.
- Use the key vocabulary.

## 1 Sort the resources. [8]

SORT THESE

**oil fish wind sunlight iron ore bamboo natural gas coal**

| Renewable | Non-renewable |
|---|---|
| | |
| | |
| | |
| | |

## 2 Put these in order from fastest to slowest to regenerate (1 = fastest). [3]

- ☐ hilsa population (1–2 years)
- ☐ a mahogany tree (20–50 years)
- ☐ soil (hundreds of years)
- ☐ natural gas (millions of years)
- ☐ bamboo (3–5 years)

## 3 Match each "R" to an example. [5]

Draw a line to match each pair, or write the letter.

| | |
|---|---|
| refuse ● | ● **A** refill a water bottle |
| reduce ● | ● **B** turn old paper into new paper |
| reuse ● | ● **C** compost vegetable peel |
| recycle ● | ● **D** turn off the tap while brushing teeth |
| rot ● | ● **E** say no to a plastic straw |

**4** **Fill in the gaps.** [3]

WORD BANK

**fossil** **sustainable** **renewable** **1971** **2002**

Using resources in a way that does not harm future generations is called _____ use.

Coal, oil and gas are _____ fuels. Bangladesh banned thin plastic bags in _____

.

# Is it sustainable?

★★☆
Grade 5

Name: Class: Date:

**Learning goals**

- Evaluate whether uses of resources are sustainable.
- Explain the impact of human activities.

**1** **For each example, decide if it is sustainable and explain why.** [8]

| Example | Sustainable? | Why? |
|---|---|---|
| Planting a new tree for every tree cut down | | |
| Catching every hilsa, even the young jatka | | |
| Using solar panels for electricity | | |
| Pumping more groundwater than the rain replaces | | |

**2** **Why is natural gas called non-renewable, even though it was made by nature?** [2]

**3** **Explain how banning plastic bags helps both the environment and people in cities.** [2]

**4** **Why does saving vultures matter to people too?** [2]

INVESTIGATE · B5 · RESOURCES

# Our class rubbish audit

★★☆
Grade 5

Name: Class: Date:

**Learning goals**

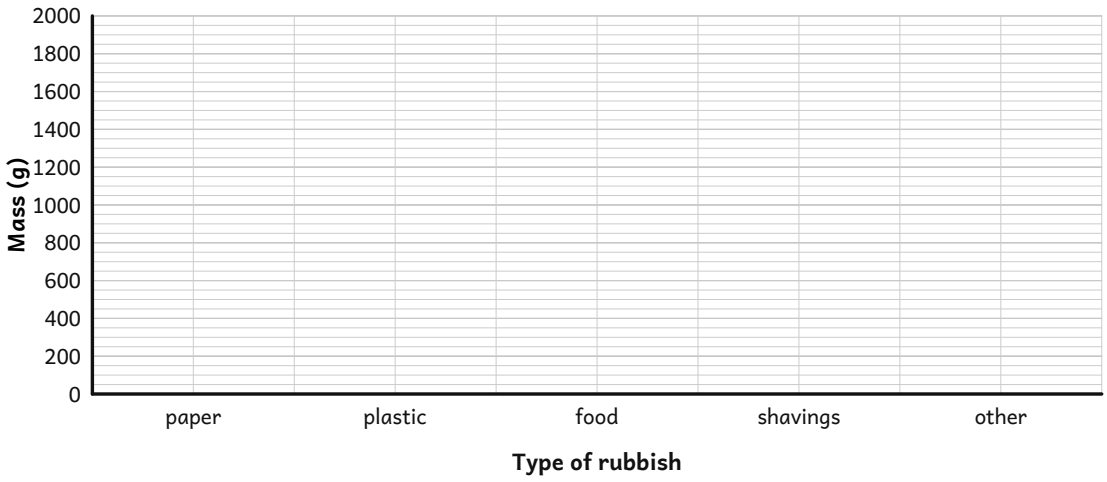
- Collect, sort and record data.
- Present data in a bar chart.
- Use evidence to plan action.

Class 5A collected all the rubbish from their classroom for one week, sorted it and weighed each type.

**1** **Complete the table.** [5]

| Type of rubbish | Mass (g) | Can it be recycled, reused or composted? |
|---|---|---|
| paper and card | 1800 | |
| plastic wrappers and bottles | 950 | |
| food waste | 1200 | |
| pencil shavings | 150 | |
| other | 300 | |

**2** **Draw a bar chart of the results.** [4]




**3** **What was the total mass of rubbish? What fraction was paper and card?** [2]

.....

.....

**4** **Use the data to suggest two actions the class could take. Say which "R" each one is.** [2]

# Saving the hilsa

★★★ Grade 5

Name: Class: Date:

**Learning goals**

- Explain how rules help a renewable resource regenerate.
- Interpret data over time.
- Consider different points of view.

Science · Economics · Maths · Ethics · Culture

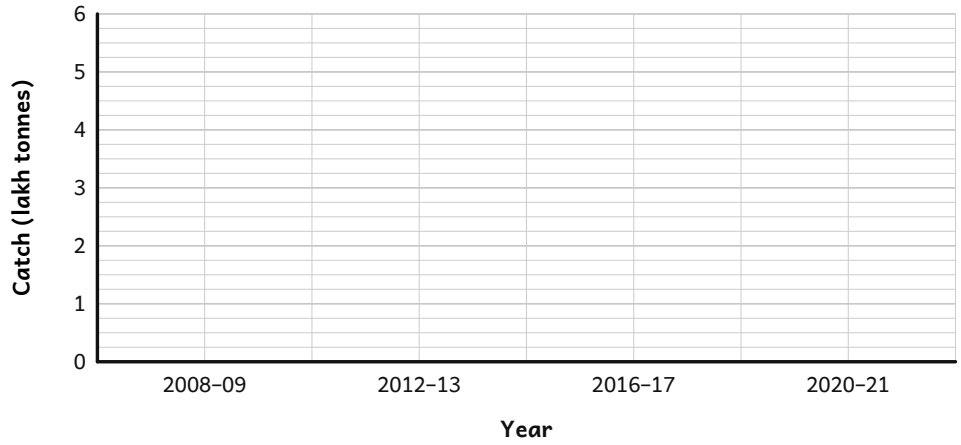
The **hilsa** (*ilish*) is the national fish of Bangladesh. Millions of people depend on catching, selling and eating it. Adult hilsa swim from the sea up rivers such as the Meghna and Padma to lay their eggs. The young fish, called **jatka**, then swim back to the sea to grow.

In the past, so many jatka and breeding adults were caught that the hilsa catch fell. So the government made rules: catching jatka is banned for several months of the year, fishing is banned for **22 days** during the main breeding season in October, and some rivers are **sanctuaries**. During the bans, poor fishing families receive free rice.

Information from the Department of Fisheries, Bangladesh. Catch figures are approximate.

**1** **Use the data (approximate hilsa catch in lakh tonnes; 1 lakh = 100 000).** [5]

| Year | 2008–09 | 2012–13 | 2016–17 | 2020–21 |
|---|---|---|---|---|
| Hilsa catch (lakh tonnes) | 3.0 | 3.5 | 5.0 | 5.7 |




**a)** Plot the graph and describe the trend.

**b)** Suggest how the fishing bans helped.

**2** **Why are the bans an example of using a renewable resource sustainably?** [2]

.....

.....

**3** **Complete the table to show different points of view.** [3]

| Person | View |
|---|---|
| a fisher during the 22-day ban | |
| a fisheries scientist | |
| a family buying fish in Dhaka | |

**4** **Suggest one more way to help fishing families during the ban.** [1]

.....

.....

# Show what you know

★★★ Grade 5

Name: Class: Date:

**Learning goals**

- Use key vocabulary.
- Answer test-style questions.

**1** **Find the key words.**

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| U | D | R | S | K | R | I | D | R | K | I |
| K | U | E | V | O | C | S | I | L | B | K |
| E | Y | D | C | G | B | W | O | S | T | G |
| Y | D | U | O | R | G | R | F | L | F | N |
| D | A | C | A | M | E | Y | E | G | A | Y |
| L | C | E | L | S | O | C | K | U | L | R |
| W | R | K | N | D | S | U | Y | R | S | T |
| G | F | O | R | E | S | T | I | C | E | E |
| I | H | I | L | S | A | G | G | R | L | W |
| R | E | N | E | W | A | B | L | E | I | E |
| R | F | C | T | E | U | Y | O | G | A | S |

**coal** **forest** **gas** **hilsa** **recycle** **reduce** **renewable** **reuse** **solar**

**2** **Which of these is a renewable source of energy?** [1]

- ◯ coal
- ◯ natural gas
- ◯ wind
- ◯ oil

**3** **What does "sustainable" mean?** [1]

- ◯ using everything now
- ◯ meeting our needs without harming future generations
- ◯ never using any resources
- ◯ only using metals

**4** **Make a poster plan to persuade your school to save one resource (water, paper or electricity). Include a fact, a picture idea and an action.** [3]

My poster plan

**5** **Challenge. Trees are renewable, yet forests can disappear. Explain how.** [2]

.....

.....

.....

**How am I doing? Colour or tick a circle.**

| I can... | Not yet | Nearly | Got it! |
|---|---|---|---|
| explain renewable and non-renewable | ○ | ○ | ○ |
| order resources by how long they take to regenerate | ○ | ○ | ○ |
| explain what sustainable means | ○ | ○ | ○ |
| give examples of the 5 Rs | ○ | ○ | ○ |
| collect and present data about waste | ○ | ○ | ○ |
| explain how fishing rules help the hilsa | ○ | ○ | ○ |